

Statistical Analysis Report:

Atmospheric Variables in Bengaluru

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1. Problem Definition

Atmospheric datasets typically contain numerous interrelated variables, such as temperature, solar radiation, and humidity. These variables often exhibit high multicollinearity, making it difficult to identify the underlying drivers of weather patterns without redundant information. The objective of this study is to apply Principal Component Analysis (PCA) to reduce the dimensionality of the Bengaluru 2019 dataset from 19 variables to a few significant, uncorrelated factors while retaining maximum variance.

2. Dataset Description

The dataset utilized for this study consists of real-world atmospheric and solar radiation data recorded in **Bengaluru, India**, for the year **2019**.

- **Location:** Bengaluru, Karnataka.
- **Temporal Scope:** Hourly/periodic observations recorded throughout the year 2019.
- **Sample Size:** The dataset contains approximately **2,000 observations** after preprocessing.
- **Number of Variables:** Originally containing multiple parameters, the study focused on **19 key variables** after excluding metadata.

3. Methodology

The analysis was conducted using R following a structured statistical workflow:

- Data Preparation: The raw dataset was filtered to remove irrelevant time-based variables (Year, Month, Day, Hour, Minute) and categorical flags (Fill Flag, Cloud Type).
- Data Cleaning: Missing values were handled using `na.omit()` to ensure a complete case analysis.
- Standardization: Because atmospheric variables use different units (e.g., Temperature in Celsius vs. Pressure in hPa), the data was standardized using Z-score normalization (`scale()`). This ensures each variable contributes equally to the analysis.
- Dimensionality Reduction: PCA was performed using the `prcomp()` function, centering and scaling the data to extract orthogonal principal components.

4. Validation of Assumptions

To ensure the dataset was suitable for PCA, the following statistical assumptions were validated:

- Sampling Adequacy (KMO Test): NULL Hypothesis: Data is not suitable for PCA (low shared variance).
 - Result: The KMO value was 0.74, which exceeds the minimum threshold of 0.60, indicating adequate shared variance for factor extraction.
- Significance of Correlation (Bartlett's Test):
 - NULL Hypothesis: Variables are not correlated (correlation matrix is an identity matrix).
 - Result: The p-value was significantly less than 0.05, allowing us to reject the null hypothesis and confirm significant correlations between variables.
- Multicollinearity: Initial correlation matrix checks showed strong relationships between solar variables (GHI, DNI, DHI), justifying the need for dimensionality reduction.

5. Statistical Analysis & Results

The PCA successfully reduced the 19 original variables into a few meaningful components.

Variance Explanation

- PC1: Explains 40.7% of the total variance.
- PC2: Explains 14.8% of the total variance.
- PC3: Explains 9.1% of the total variance.
- Cumulative: The first three components account for 64.7% of the total information in the dataset.

Component Interpretation (Loadings)

By examining the variable loadings, the components are interpreted as follows:

1. PC1 (Solar Radiation Factor): High positive loadings for GHI, DNI, DHI, and Temperature. This component represents solar energy intensity.
2. PC2 (Moisture & Atmospheric Factor): High loadings for Precipitable Water, Dew Point, and Aerosol Optical Depth. This represents air composition and moisture levels.
3. PC3 (Wind & Pressure Factor): High loadings for Pressure and Wind Speed, representing atmospheric dynamics.

6. Interpretation & Conclusion

The analysis demonstrates that weather patterns in Bengaluru can be simplified into three primary dimensions: Solar Intensity, Atmospheric Moisture, and Dynamic Pressure.

The Biplot reveals that solar radiation variables point in the same direction, indicating strong positive correlation, while humidity and zenith angle lie in the opposite direction, indicating an inverse relationship. By reducing the dataset from 19 to 3 components, we have simplified the complex atmospheric data while retaining nearly 65% of the original variance, providing a more efficient model for environmental forecasting.